

	A	SP
2023 Q1	100	0.875
Q2	120	1.04
Q3	110	0.958
Q4	130	1.125
2024 Q1	110	0.875
Q2	130	1.04
Q3	120	0.958
Q4	140	1.125

Forecast $\begin{cases} L+T+S \\ (L+T) \times S \end{cases}$

$$A = S \times t$$

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$$\frac{A}{S}$$

$$\alpha = 0.5$$

$$\beta = 0.5$$

H. Method

Sales	S_t	t_t	F_t
80	80	0	80
82	81	0.5	81.5
84	82.75	1.125	83.875
89	86.437	2.4062	88.843
90	89.421	2.6953	92.117
91	91.558	2.916	93.974
95	94.487	2.6723	97.159
98	97.579	2.8824	100.46
99	99.731	2.5168	102.24

$$S_t = \alpha A_t + (1-\alpha)(S_{t-1} + T_{t-1})$$

$$T_t = \beta (S_t - S_{t-1}) + (1-\beta)T_{t-1}$$

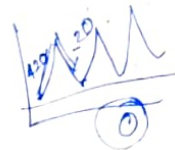
$$F_{t+1} = S_t + T_t$$

H.W.	Q1	Q2	Q3	Q4
2024	100	120	110	130
2025	110	130	120	140

$$T \cdot A = 120$$

total/avg

	0.87	1.04	0.95	1.12
SPM	0.87	1.04	0.95	1.12
SPA	-15	5	-5	15



$$Q1 = \frac{100+110}{2} = 105$$

$$SPQ1 = 105 \div 120$$

$$Q2 = \frac{120+130}{2} = 125$$

$$SP2 = 125 \div 120 =$$

$$Q3 = \frac{110+120}{2} = 115$$

$$SP3 = 115 \div 120 =$$

$$Q4 = \frac{130+140}{2} = 135$$

$$SP4 = 135 \div 120 =$$

$$SP1 = 105 \div 120 =$$

$$SP2 = 125 \div 120 =$$

$$SP3 = 115 \div 120 =$$

$$SP4 = 135 \div 120 = 15$$

$$L_t = 140 - SP(Q1) = 140 -$$

$$t = L_t - 120 (\text{cov. avg})$$

$$F_t + A = L_t + t_t + SP$$

$$F_{t+1} = (L_t + t_t) \cdot SP_{Q1}$$

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$$\hat{y} = -235(t) + 22860$$

t y_t

1 22600

2 22400

3 22250

4 21800

5 21725

$$\hat{y} = -235(1) + 22860 = 22625$$

$$\hat{y} = -235(2) + 22860 = 22390$$

$$\hat{y} = -235(3) + 22860 = 22155$$

$$\hat{y} = -235(4) + 22860 = 21920$$

$$\hat{y} = -235(5) + 22860 = 21685$$

$$\text{MAPE} = \frac{100}{n} \sum_{t=1}^n \frac{|y_t - \hat{y}_t|}{y_t}$$

$$t=1 \Rightarrow \frac{100}{5} \sum_{t=1}^n \left(\frac{22600 - 22625}{22600} \right) = \frac{100}{5} \left(\frac{25}{22600} \right) = 0.0221$$

t	y _t	$\hat{y}_t$	$y_t - \hat{y}_t$	$(y_t - \hat{y}_t)^2$	MAPE	MAD	MSE
1	22600	22625	-25	625	0.0221	5	125
2	22400	22390	10	100	0.0089	2	20
3	22250	22155	95	9025	0.0853	19	1805
4	21800	21920	-120	14400	0.1100	24	2880
5	21725	21685	40	1600	0.0368	8	320

$$t=2 \Rightarrow \frac{100}{5} \sum_{t=1}^n \frac{10}{22400} = 0.0089$$

$$t=3 \Rightarrow \frac{100}{5} \sum_{t=1}^n \frac{95}{22250} = 0.0853$$

$$t=4 \Rightarrow \frac{100}{5} \sum_{t=1}^n \frac{120}{21800} = 0.1100$$

$$t=5 \Rightarrow \frac{100}{5} \sum_{t=1}^n \frac{40}{21725} = 0.0368$$

$$\text{MAD} = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t|$$

$$t=1 \Rightarrow \frac{1}{5} (25) = 5$$

$$t=2 \Rightarrow \frac{1}{5} (10) = 2$$

$$t=3 \Rightarrow \frac{1}{5} (95) = 19$$

$$t=4 \Rightarrow \frac{1}{5} (120) = 24$$

$$t=5 \Rightarrow \frac{1}{5} (40) = 8$$

$$\text{MSE} = \frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2$$

$$t=1 \Rightarrow \frac{1}{5} (625) = 125$$

$$t=2 \Rightarrow \frac{1}{5} (100) = 20$$

$$t=3 \Rightarrow \frac{1}{5} (9025) = 1805$$

$$t=4 \Rightarrow \frac{1}{5} (14400) = 2880$$

$$t=5 \Rightarrow \frac{1}{5} (1600) = 320$$

t	y _t	$\hat{y}_t$	$y_t - \hat{y}_t$	$(y_t - \hat{y}_t)^2$	MAPE	MAD	MSE	R ²
1	22600	22625	-25	625	0.0221	5	125	
2	22400	22390	10	100	0.0089	2	20	
3	22250	22155	95	9025	0.0853	19	1805	
4	21800	21920	-120	14400	0.1100	24	2880	
5	21725	21685	40	1600	0.0368	8	320	
					0.2608	58	5150	

y_t = 22155

$$R^2 = 1 - \frac{\sum_{t=1}^n (y_t - \hat{y}_t)^2}{\sum_{t=1}^n (y_t - \bar{y}_t)^2}$$

$$= 1 - \frac{25725}{22155}$$

23/09/26

$$Y_t = \{y_1, y_2, y_3, \dots, y_t\}$$

where $t = 1, 2, 3, \dots, t$

AR process: Relates the current value of its own past values (lags)

$$1.25579 + 1.084278 Y_{t-1}$$

$$Y_t \quad Y_{t-1} \quad e(Y_t - \hat{Y}_t)$$

$$1 \quad -$$

$$3 \quad 1$$

$$5 \quad 3$$

$$6 \quad 5$$

$$8 \quad 6$$

$$10 \quad 8$$

$$11 \quad 10$$

$$12 \quad 11$$

$$15 \quad 12$$

$$18 \quad 15$$

$$20 \quad 18$$

$$24 \quad 20$$

$$Y_t = \alpha_0 + \alpha_1 Y_{t-1} + \alpha_2 Y_{t-2} + \dots + \alpha_p Y_{t-p} + \varepsilon_t$$

$$\varepsilon_t = 1 + \beta_1 \varepsilon_{t-1} + \beta_2 \varepsilon_{t-2} + \dots + \beta_q \varepsilon_{t-q} + \varepsilon_t$$

$$ARMA(1,0)$$

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$$AR(p) \Rightarrow Y_t = \alpha_0 + \alpha_1 Y_{t-1} + \alpha_2 Y_{t-2} + \dots + \alpha_p Y_{t-p} + \varepsilon_t$$

$$Y_t = \alpha_0 + \sum_{i=1}^p \alpha_i Y_{t-i} + \varepsilon_t$$

Moving avg: MA relates the current value to past white noise error term.

$$AR \Rightarrow Y_t \rightarrow f(Y_{t-1})$$

$$MA \Rightarrow \varepsilon_t = f(\varepsilon_{t-1})$$

ARMA: Merges both concepts as future points based on previous demands w/ shocks

$$MA(2) \Rightarrow Y_t = \mu_0 + \mu_1 \varepsilon_{t-1} + \mu_2 \varepsilon_{t-2} + \varepsilon_t$$

$$MA(q) \Rightarrow Y_t = \mu_0 + \mu_1 \varepsilon_{t-1} + \dots + \mu_q \varepsilon_{t-q} + \varepsilon_t$$

$$MA(q) \Rightarrow Y_t = \mu_0 + \sum_{i=1}^q \mu_i \varepsilon_{t-i} + \varepsilon_t$$

$$ARMA(1,1) \Rightarrow Y_t = \alpha_0 + \alpha_1 Y_{t-1} + \mu_1 \varepsilon_{t-1} + \varepsilon_t$$

$$ARMA(1,2) \Rightarrow \alpha_0 + \alpha_1 Y_{t-1} + \alpha_2 Y_{t-2} + \mu_1 \varepsilon_{t-1} + \mu_2 \varepsilon_{t-2} + \varepsilon_t$$

$$ARMA(0,0)$$

$$25/09/26$$

$$ARMA(1,1) \Rightarrow Y_t = 10 + 0.6 Y_{t-1} + \varepsilon_t + 0.4 \varepsilon_{t-1} + \varepsilon_t$$

$$Y_t = 10 + 0.6(20) + 0.4(5) + 0$$

$$Y_t = 24$$